DATA VISUALIZATION COURSE PROJECT · PROCESS BOOK

OLYMPIC SUCCESS OF COUNTRIES – MORE THAN JUST MEDALS?

The medal table rewards the big and the rich. We adjust for wealth and population to find the countries that truly outperform their expectations.

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(alphabetical order)

EPFL · COM-480 · MILESTONE 3

01 INTRODUCTION

HOW DID THE IDEA START?

We wanted a dataset that supports **strong visuals** and a **clear data-driven story**. The Olympics were a natural fit, with one central question: when countries win many medals, how much is sporting merit and how much reflects population and economic resources? We combined an **Olympic medals dataset (1896–2024, Kaggle)** with **World Bank GDP and GDP-per-capita data**, focusing on 1960 onward where both sources overlap.

Prior work[1] shows a positive relationship between wealth and Olympic success, and Fig. 1 reflects this: countries with larger GDPs generally win more medals. We then ask what lies beyond this simple correlation: are overachievers mainly rich countries, and are underachievers mainly less wealthy ones?

The key modeling choice was to use **relative performance** instead of raw totals. We estimate how many medals a country is *expected* to win from population and GDP per capita, then measure deviations from that baseline. This gives the project a consistent narrative and makes results interpretable for sports fans, economists, and readers interested in fairness.

PRE-PROCESSING OF DATA

The sources do not align directly: Olympic data uses **IOC country codes**, while World Bank data uses ISO3 codes. We reconciled them with a mapping table (*utils.code_fix*), including historical entities such as East/West Germany, the USSR, and Czechoslovakia, mapped to modern successors. We then aggregated medals at country-year-season level and merged with GDP by code and year.

Because World Bank series do not provide population directly, we derived it as **total GDP / GDP per capita**. After filtering missing and non-positive values, we pre-computed expected-medals model outputs in Python so the website serves lightweight, ready-to-render data.

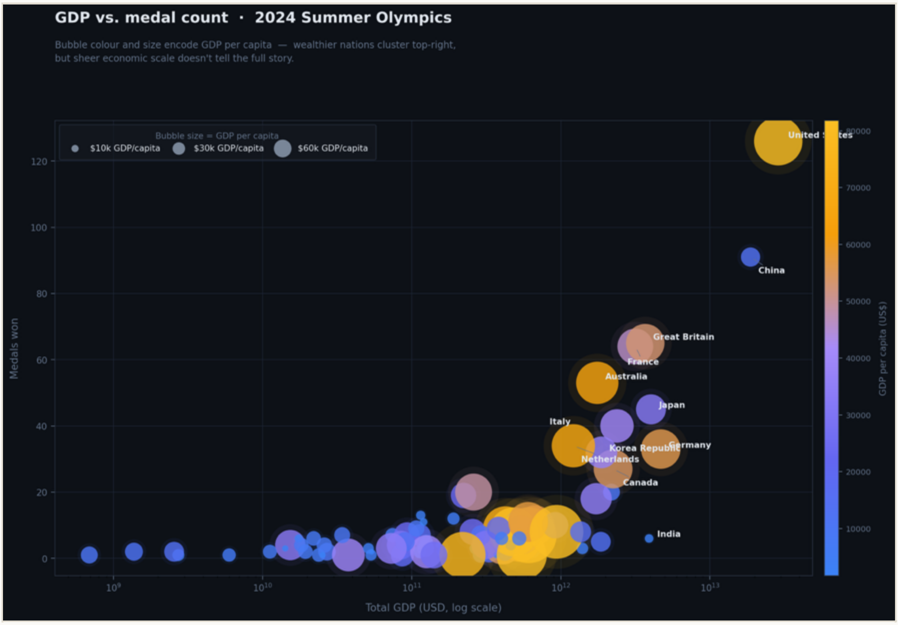


Fig. 1. *Total GDP vs. medals won, 2024 Summer Olympics. Bubble size and colour encode GDP per capita.*

[1] Bernard & Busse (2004), *Review of Economics and Statistics*, 86(1), 413–417.

02 VISUALIZATIONS

Based on our story and on the features surfaced by our exploratory analysis, we designed four visualizations. They form a single scrolling narrative on the website, moving from the familiar total medal count toward our resource-adjusted view of success.

DUAL INTERACTIVE GLOBE

Two synchronized 3-D globes side by side. The first colours countries by total Olympic medal count; on scroll, a second globe slides in colouring them by relative performance instead. Both can be dragged to rotate, and hovering a country reveals its values.

> **GOAL** to confront the traditional medal-count worldview with our resource-adjusted one, and let the viewer immediately see how the picture of “successful” countries changes. It intentionally leaves the computation of "relative performance" open to create some tension and curiosity on the side of the reader.

OUR METHODOLOGY

An explanatory section with an animated 3-D regression plane showing how expected medals are fit from population and GDP per capita, making the model behind the rest of the story transparent before we going into depth analysis.

> **GOAL** to explain, visually and honestly, exactly how we compute “expected” medals so the over- and under-performance results that follow are interpretable and trustworthy.

RELATIVE-PERFORMANCE SCATTER (GAPMINDER-STYLE)

An interactive bubble chart with GDP per capita on a log x-axis and the actual-to-expected medal ratio on the y-axis. Bubble size encodes medals won, colour encodes the z-score of over- or under-performance, a season toggle switches Summer/Winter, and a year slider with a play button animates through Olympic history. Clicking a bubble pins a country and traces its trajectory, while a side leaderboard lists the top over- and under-performers.

> **GOAL** to let users explore, per Games, which countries punch above their economic weight and which fall short, and to follow individual countries as they drift relative to expectation across the decades.

OLYMPIC EFFICIENCY OVER TIME (BAR CHART RACE)

Two synchronized racing bar charts. The left race ranks countries by cumulative total medals since 1960; the right ranks the same timeline by cumulative efficiency, defined as cumulative actual medals divided by cumulative expected medals. A value of 1 means exactly as expected, while 2 means twice as many medals as expected. Both advance in lockstep, include a seek slider, and a season toggle switches between Summer and Winter Games.

> **GOAL** to contrast the two ways of telling the story over time — who simply wins the most versus who performs best for their resources — and reveal how that ranking shifts across Olympic history.

03 IMPLEMENTATION DETAILS

After settling on the visualizations, we built the website as a single-page scrolling experience (tested on Chrome and Safari). For the second milestone, we had a first running version with the basic skeleton and first drafts of the charts, which we completed into the full interactive version for the third. We first sketched each visualization on paper and with a digital stylus during team meetings, then implemented them in the browser — the charts are hand-built directly with SVG inside React components, complemented by a few specialised libraries.

Main tech stack

React + TypeScript

The whole site is a React single-page app in TypeScript for type safety and reusable, maintainable components.

Vite

A fast modern build tool for development and bundling, giving instant live reload while iterating on the charts.

WebGL / SVG

The two globes are rendered in WebGL via *react-globe.gl* (Three.js under the hood), while the scatter plot and methodology charts are drawn directly as SVG with custom scales and log axes.

Specialised libraries

`racing-bars` (the bar chart race) and framer-motion (layout animation).

Python · pandas / NumPy

Milestone-1 analysis and all pre-processing: code reconciliation, merging, deriving population, and fitting the per-event regressions exported to the site.

GitHub Pages

We host the site on GitHub Pages — permanently online with no server to maintain. Deployment is a single **npm run deploy** that builds and pushes to the gh-pages branch.

Reusable building blocks

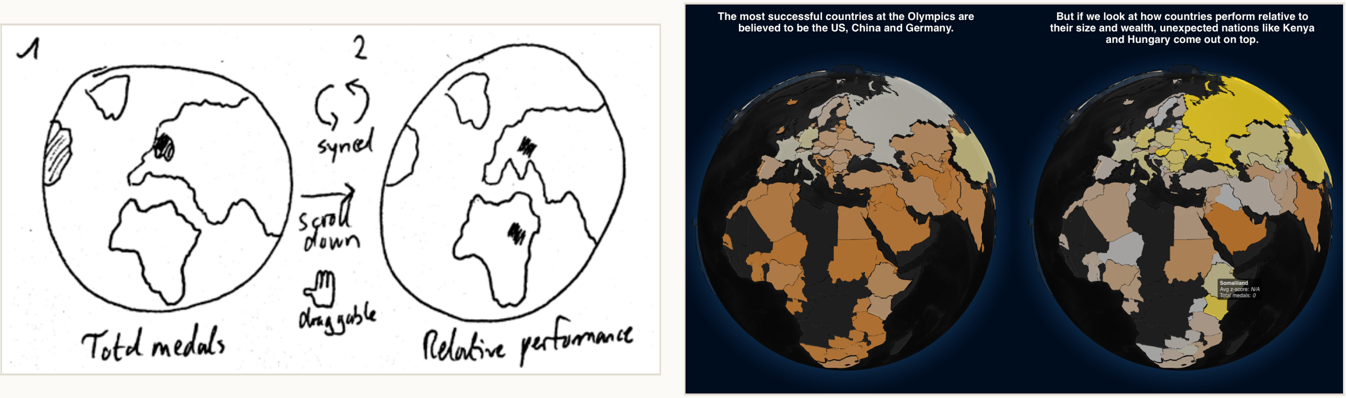
To follow good practice and reduce duplication, we shared logic across the visualizations. The same **season toggle and animated time controls** — with a play button and seek slider — appear in both the scatter plot and the bar chart race, and a single **IOC-to-flag mapping** drives the country flags throughout. The regression model is computed once in Python and consumed identically by every chart, so the whole site tells one internally consistent story.

04 EXPLAINING THE VISUALIZATIONS

DUAL INTERACTIVE GLOBE

The globe is the entry point of the website: it takes the most familiar metric in Olympic reporting — medal counts by country — and immediately questions it. The first globe colours countries on a bronze-to-gold scale by total medals, with each country's height scaled to its haul, so the usual suspects (the United States, China, Germany) dominate. As the user scrolls, the view **splits**: the medal globe slides left and a second globe slides in, coloured by **relative performance** (average z-score across all Olympic Games). On that second globe very different countries light up — Kenya and Hungary among them — setting up the central tension of the project. The two globes are camera-synchronized, so the viewer always compares the same hemisphere under both definitions of success; we used the *react-globe.gl* WebGL library for performant rendering.

SKETCH → IMPLEMENTATION



Side-by-side globes contrasting medal count (left globes) with relative performance (right globes).

OUR METHODOLOGY

Because the rest of the story depends on the idea of "expected" medals, we make the model transparent. We fit a **log-linear regression** separately for each Olympic event and season:

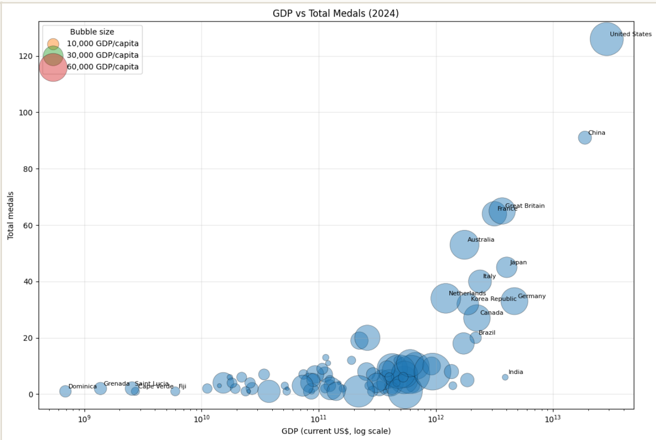
$$\log(medals) = \alpha \cdot \log(GDP\ per\ capita) + \beta \cdot \log(population) + \gamma$$

Fitting per event controls for differences in population and wealth at that point in time, so the residual measures genuine over- or under-performance rather than era effects. Including population as well as GDP per capita matters: a large, rich country like the United States gets a high baseline expectation and a z-score near zero, while small countries that consistently exceed their predicted medals — Jamaica, Kenya, Cuba — emerge as the real outliers. We express over-performance multiplicatively, as **log(actual / predicted) ÷ residual standard deviation**. On the website this is conveyed through an animated 3-D regression plane that sweeps in to show the expected surface.

RELATIVE-PERFORMANCE SCATTER (GAPMINDER-STYLE)

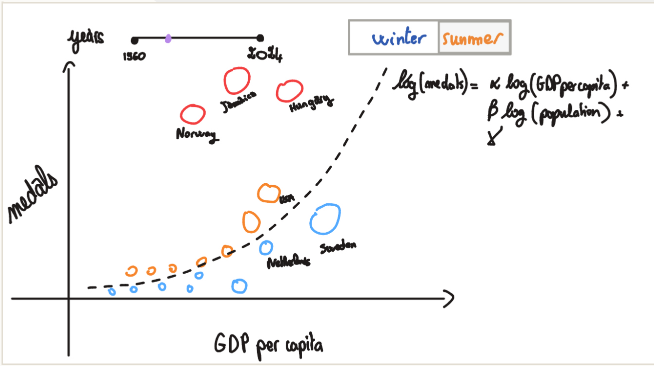
This is the analytical heart of the project and the visualization that evolved most between milestones. Our milestone-1 exploration started from a simple bubble chart of GDP versus total medals, which already hinted that wealth and medals move together — but imperfectly.

EXPLORATION



M1: total medals vs GDP, bubble size = GDP per capita.

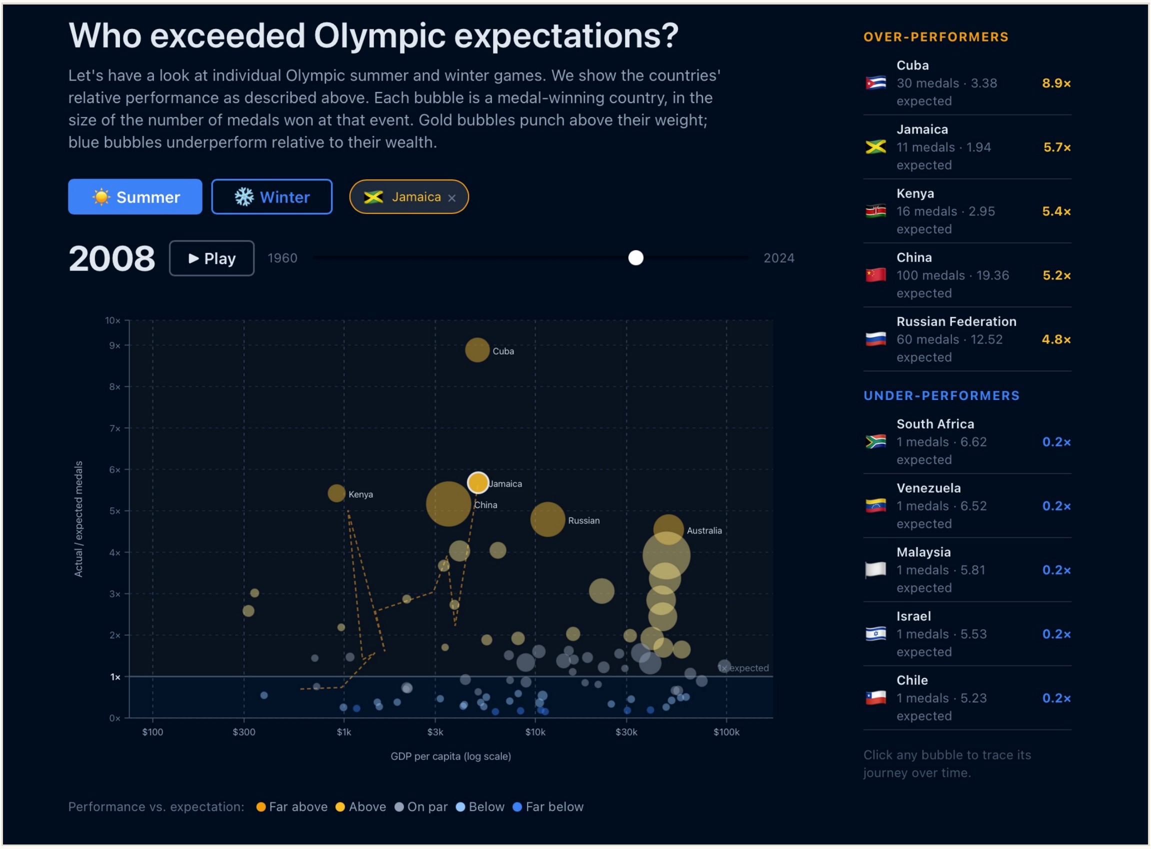
SKETCH



M2 sketch of the efficiency scatter.

By milestone 3 it had become a fully interactive **Gapminder-style** chart. Each bubble is a medal-winning country at a given Games. The x-axis is GDP per capita on a log scale; the y-axis is the **actual-to-expected medal ratio**, with a clear baseline at “1× expected”. Bubble size encodes medals won, colour encodes the z-score on a diverging scale — gold for far above expectation, blue for far below. A **season toggle** switches the dataset, a **year slider with a play button** animates through every edition, and **clicking a bubble pins that country**, drawing a dashed trajectory through previous Games. A side leaderboard lists the top over- and under-performers.

FINAL IMPLEMENTATION

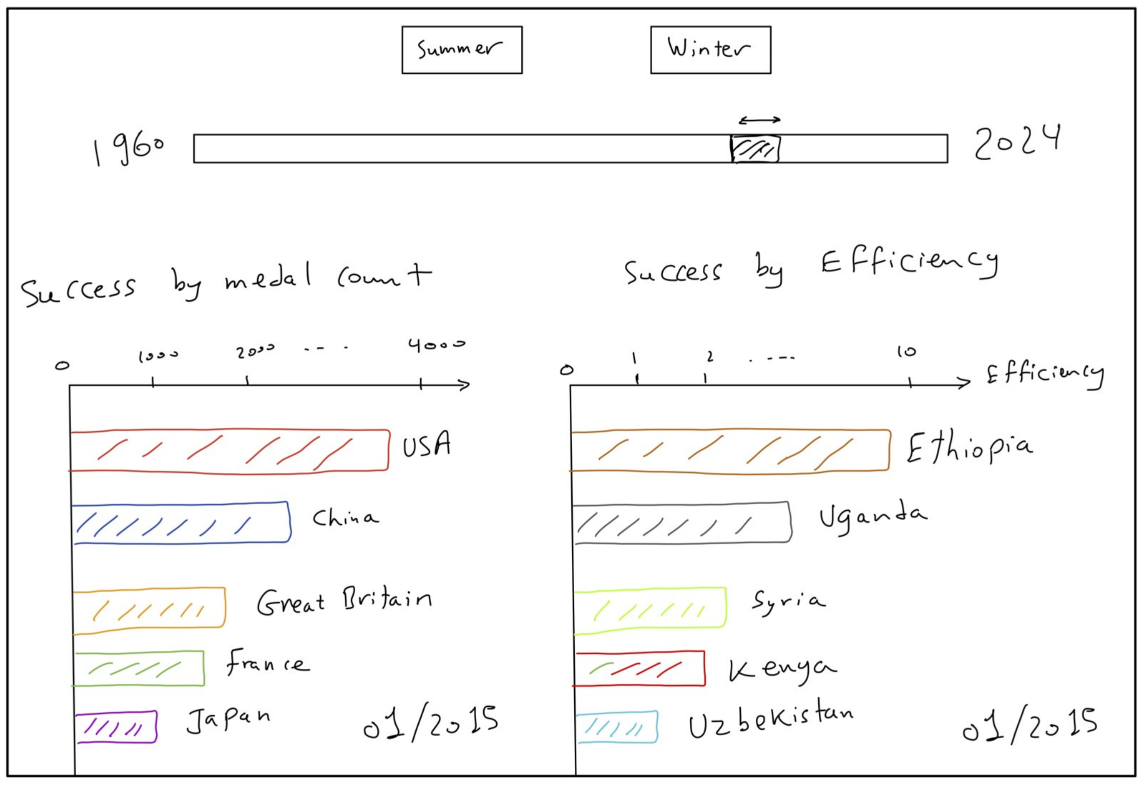


The deployed relative-performance scatter — here pinning Jamaica (2008), with the over/under-performer leaderboard.

OLYMPIC EFFICIENCY OVER TIME (BAR CHART RACE)

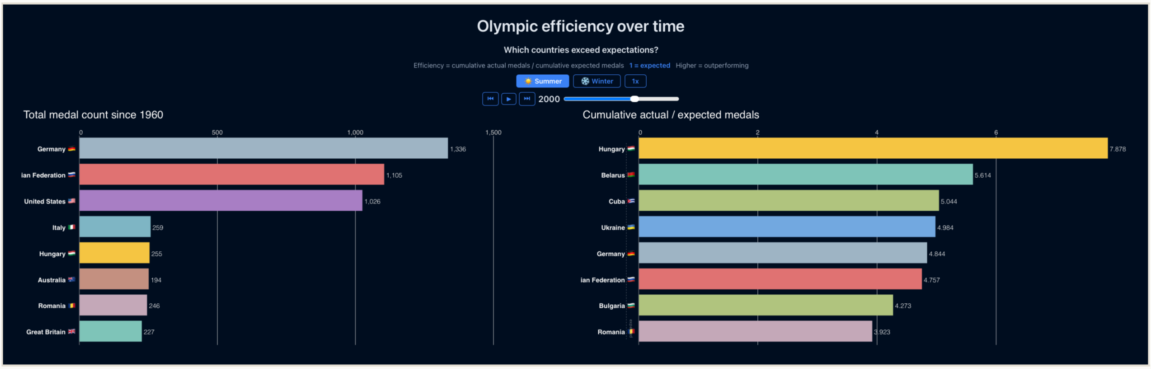
The final visualization brings time to the foreground by racing two leaderboards side by side. The **left race** ranks countries by cumulative total medals since 1960. The **right race** ranks countries by cumulative efficiency. For each Olympic year, we estimate how many medals a country would be expected to win from its population and GDP per capita. We then add actual medals and expected medals from 1960 up to the year currently shown, and compute **cumulative efficiency = cumulative actual medals / cumulative expected medals**. A value of 1 means exactly as expected; 2 means twice as many medals as expected; 0.5 means half as many as expected.

SKETCH



Initial sketch for comparing raw Olympic success with resource-adjusted performance over time.

FINAL IMPLEMENTATION



The final chart race compares cumulative medal totals on the left with cumulative efficiency on the right.

We implemented the race with [racing-bars](). Each bar is one country (with flag). Left x-axis: cumulative medals since 1960. Right x-axis: cumulative efficiency ratio (actual/expected), so longer bars mean stronger over-performance. The year label marks the current Olympic year. Controls are synchronized across both panels (season toggle, play/pause, step, speed and timeline slider). Historical teams are merged into modern country codes before modeling.

05 CHALLENGES

Our team had solid web-development experience, so the main challenges were reconciling two different datasets and making fair comparisons across countries and eras.

1. **Reconciling country codes across sources and history.** The datasets use different codes, and history adds complexity: dissolved or split nations (USSR, East/West Germany, Czechoslovakia, Yugoslavia) have no single modern GDP series. We mapped them to successor states and merged rows when multiple historical teams share one modern code, avoiding double-counting.
2. **Defining a fair success metric.** Raw medal counts mostly track economic size, so total rankings favor big, rich countries. Our expected-medals model (fit per event in log space with population and GDP per capita) surfaces small countries that genuinely over-perform.
3. **Deriving population from the available data.** The World Bank extracts had no population field, but the model needs it. We derived it as total GDP / GDP per capita, then filtered non-positive or missing artefacts.
4. **Usability of the website.** Complex interactions, especially on the globes, were not always obvious, so we added hints like "Drag to explore". We also enforced a roughly 16:9 screen ratio to keep layout and proportions consistent.

06 PEER ASSESSMENT

All members contributed across the project, and most design and implementation decisions were made together. Beyond that shared work, each member focused on specific areas, reflected in the commit history.

Fares Fawzi

RACING CHARTS · EFFICIENCY METRICS · EDA

Implemented the racing bar charts and efficiency calculations. Led core EDA and dataset preparation (including medals-GDP merge and GDP rank-by-year). Fixed duplicate-country and layout bugs, and polished the codebase and README for MS3.

Dominik Glandorf

ARCHITECTURE · GLOBES · DEPLOYMENT

Set up the website scaffold and GitHub Pages pipeline (Vite, React, TS). Main author of the page structure, dual-globe visualization, and global styling. Drove sport-level analysis and maintained the M1 report and README.

Mohamed Mamouri

EFFICIENCY CONCEPT · SCATTER · PROCESS BOOK

Originated the efficiency/expected-medals framing and bubble-chart design behind the relative-performance scatter, and built the component with pre-computed data. Produced early medals-vs-GDP analysis and the M2 sketch, contributed to the README, and wrote this process book.

May this resource-adjusted view make the Games a fairer contest to read — and reveal the quiet overachievers the medal table tends to hide.